

Linux Kernel Commands - Cheat Sheet

uname -r (Unix Name)

Displays the running kernel version.

dmesg (Display Message)

Shows kernel ring buffer logs (boot, drivers, kernel messages).

lsmod (List Modules)

Lists currently loaded kernel modules.

modprobe (Module Probe)

Loads/removes modules with dependencies.

insmod (Insert Module)

Inserts a kernel module (.ko) manually.

rmmod (Remove Module)

Removes a kernel module manually.

depmod (Dependency Module)

Generates module dependency list.

modinfo (Module Info)

Shows information about a .ko module.

make menuconfig (-)

Text UI for configuring kernel features and drivers.

make bzImage (Big zImage)

Builds a compressed kernel binary image.

make modules (-)

Builds kernel modules from source.

make modules_install (-)

Installs built modules to /lib/modules.

make install (-)

Installs kernel and updates the bootloader.

kexec (Kernel Exec)

Boots a new kernel without rebooting.

sysctl (System Control)

Views or changes kernel runtime parameters.

top / htop (-)

Shows real-time process/kernel thread activity.

ps -eLf (-)

Displays kernel threads with full command-line.

cat /proc/version (-)

Shows kernel version and build info.

cat /proc/modules (-)

Displays currently loaded modules.

cat /proc/kallsyms (-)

Displays kernel symbol table.

nm vmlinux (Name Mangling)

Lists all symbols in the kernel image.

gdb vmlinux (-)

Debugs compiled kernel image.

objdump -d vmlinux (-)

Disassembles kernel binary to assembly.

strace (-)

Traces system calls made by processes.

ftrace (-)

Kernel function and syscall tracer.

perf (-)

Traces CPU cycles, kernel events.

trace-cmd (-)

Frontend for ftrace tracing tool.